1. Organisms and their Environment

1.1. Energy Flow

- The sun is the principal source of energy input to biological systems.
- Energy flow is **NOT** a cycle; it starts from the sun, and then that energy is harnessed by plants, which are eaten by animals, which other animals eat.
- At each step, energy is lost to the environment.

Food Chains and Food Webs

Food Chain: a chart showing the flow of energy (food) from one organism to the next, beginning with a producer, for example:

Food Web: showing a network of interconnected food chains.

- Energy is transferred between organisms in a food chain by ingestion
- **Producer:** an organism that makes its organic nutrients, usually using energy from sunlight through photosynthesis
- **Consumer:** an organism that gets its energy by feeding on other organisms.
  - Consumers may be classed as **primary, secondary, tertiary and quaternary** according to their position in a food chain
- **Herbivore:** an animal that gets its energy by eating plants
- **Carnivore:** an animal that gets its energy by eating other animals
- **Decomposer:** an organism that gets its energy from dead or waste organic matter (i.e. a saprotroph)
- **Trophic level:** the position of an organism in a food chain, food web or ecological pyramid.
  - **Primary consumer:** eat vegetables
  - **Secondary consumer:** eat meat/drink milk
  - **Tertiary consumer:** eat a predatory fish, salmon

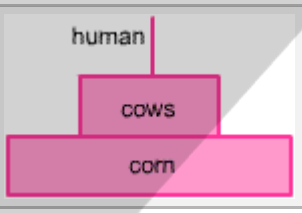
Food chains usually have fewer than five trophic levels because energy transfer is inefficient:

- Sun produces light, and less than 1% of the energy falls onto leaves.
- Producers 'fix' only about 5-8% of that energy because of transmission, reflection and incorrect wavelength.
- Primary consumers only get between 5-10% because some parts are indigestible (e.g., cellulose) and do not eat the whole plant.
- The secondary consumer gets between 10-20% because the animal matter is more digestible & has a higher energy value.
- At each level, heat is lost by respiration.

Humans eating plants is more efficient than humans eating animals because:

- We need only a couple of vegetables to have one meal but to have the meat, we must feed the animal a lot of plant material to get far less meat.
- When raising an animal, plants lose energy to the environment. Then, the animal loses energy to the environment and does not use up all the plant material, so it is inefficient.

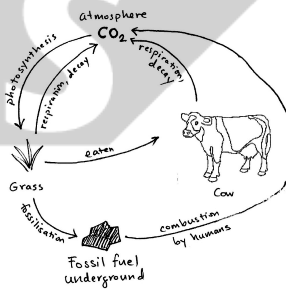
Ecological Pyramids

Pyramid of Numbers	Pyramid of Biomass
	
Shows the number of each organism in a food chain	Pyramid, which shows the biomass
When moving up the pyramid, the number of individuals decreases	(number of individuals × their individual mass)

The pyramids of biomass are **always** pyramid-shaped or upright.

1.2. Nutrient Cycles

Carbon Cycle



- Carbon is taken from the atmosphere by photosynthesis (plants)
- It is passed on to animals and decomposers by feeding.
- It is returned by respiration in plants and animals and decomposed by microorganisms.

Fossilisation is **NOT** needed anymore - from 2023 onwards

Nitrogen Cycle

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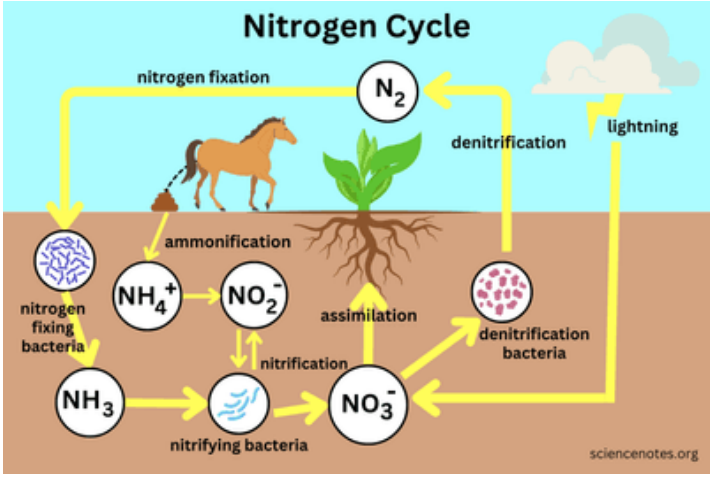


- **Nitrogen-fixing bacteria** provide usable nitrogen for plants; these may exist in the root nodules where they live in symbiosis with the plants (nitrogen fixation), or this can happen because of lightning or microorganisms providing them through decomposition.
- **Nitrifying bacteria** convert nitrogen-containing substances into nitrates for the plants (nitrification).
- Plants absorb these substances and convert them into proteins
- Death and decay happen at each trophic level, leading to stage one
- **Denitrifying bacteria** carry out denitrification: they convert nitrogen-containing substances into atmospheric nitrogen

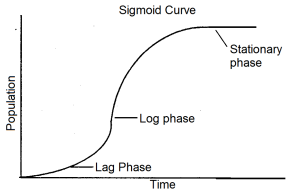
1.3. Population

- **Population:** a group of organisms of one species living in the same area at the same time.
- **Community:** all of the populations of different species in an ecosystem.
- **Ecosystem:** a unit containing the community of organisms and their environment interacting together.

Factors Affecting the Rate of Population Growth

- **Food Supply:** quantity and quality; snails need calcium to reproduce to make a shell.
- **Predation:** if the predator population falls, the prey population will rise.
- **Disease:** causes organisms to die, so a high death rate partly cancels out the birth rate, meaning less population growth, especially if the organism dies before giving birth, or even population decline.
- **Natural Disasters or Environmental Changes**: Droughts, floods, or habitat destruction can reduce population size suddenly
- **Immigration and Emigration**: People moving in and out of the population

1.4. Sigmoid Curve of Population Growth



- **Lag phase:** The number of mature, reproducing individuals is low and they may be widely dispersed
- **Exponential (Log) phase:** exponential growth occurs, the conditions are ideal, and the maximum growth rate is reached. Limiting factors do not limit growth much.
- **Stationary phase:** limiting factors slow growth as the population has reached the "carrying capacity" of its environment; when mortality rate = birth rate, the curve levels off and fluctuates around this maximum population size.
- **Death phase**: death rate > birth rate due to lack of food, competition, etc.

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